Muppidi Sai Rajeshwara Siddarth Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.56
Sri Chaitanya	Class XII (TSBIE)	2023	98.30%
Sri Velocity High School	Class X (SSC)	2021	10.00%

SCHOLASTIC ACHIEVEMENTS

- **Awarded Excellence in Academics** by Director of IITH; maintained **Department Rank 1** across 4 consecutive semesters.
- Secured **Rank 22** in **ICPC Asia West Finals 2026** among 90 top teams; advanced via **AIR 26** Kanpur & **AIR 40** Amritapuri.
- Achieved a peak rating of **1960**, hitting **Candidate Master** on Codeforces: SiddarthMSR
- Achieved 6 stars with **max rating 2233** and **ranked 359 globally** on CodeChef: sddarth msr.
- Secured **Global Rank 6** in CodeChef Starters 192.
- Secured **Rank 144** in **IICPC QuantFest Regionals** 2026 among top 600 paritcipants who advanced to regionals.
- Secured **Rank 350** in **IICPC QuantFest** 2026 prelims with over **11,000+** participants, advanced to Regionals.

WORK EXPERIENCE

Quadeye Securities LLP. | *Quantitative Strategist Intern*

- Built a modular, config-driven evaluation pipeline for **masked time-series target prediction**, automating feature engineering, model training, **hyperparameter optimization**, out-of-sample backtesting, and automated dashboard reporting.
- Executed an end-to-end **feature addition** strategy for **XGBoost models** to enhance out-of-sample generalization, engineering **temporal and relational features** (lags, rolling statistics, cross-correlations) to isolate high-signal predictive inputs while mitigating **feature-expansion overfitting**.
- Devised a custom **validation-aware SHAP feature selector** using a **4-slice expanding window** and bucketed Pearson correlation objective, successfully compressing the **in-sample** vs. **out-of-sample metric gap**
- Boosted **out-of-sample Pearson correlation** from **-0.17% to +1.26%** on Target 1 (flipping signal to positive) and **10.74% to 12.14%** on Target 2, while strictly maintaining Spearman/Kendall rank metrics

RELEVANT COURSES

- **Mathematics:** Convex Optimization (A), Linear Algebra (A+), Probability Theory (A), Differential Equations (A+), Applied Statistics, Vector Calculus (A+), Calculus I & II (A/A+), Predictive Modeling (A+), Bayesian Statistics (A)
- **Computer Science:** Algorithms (A), Data Structures & Applications (A / Lab A+), Theory of Computation (A), DBMS (A), Compilers-I (A), OS-1 (A-), NLP (A), Computer Networks (Ongoing)

SKILLS

- **Programming & Scripting Languages**: C, C++, Python, SQL, Git
- **Tools & Libraries**: Pandas, NumPy, Scikit-learn, OpenCV, PyTorch, HuggingFace, Latex

PROJECTS

Approximation Algorithms and Simulated Annealing | *DSA Lab Project* | *GitHub*

- Implemented exact, approximation, and **simulated annealing algorithms** for NP-hard problems including **Vertex Cover, Set Cover, TSP, Subset Sum, and scheduling** (DSA Lab Project under Prof. Vikas Krishnamurthy).
- Developed a **simulated annealing framework** with **probabilistic state acceptance**, **temperature cooling**, and **local neighborhood perturbation** to optimize **combinatorial search spaces for graph and set-based problems**.
- Benchmarked performance to analyze trade-offs across **runtime efficiency, memory overhead, and solution optimality**.

Age-Related Macular Degeneration (AMD) Onset Prediction | *GitHub*

- Developed a **temporal deep learning** pipeline on longitudinal **Electronic Health Records** (EHR) from 20k+ patients (5.6k cases, 14.3k controls) to predict AMD onset across **6 clinical horizons** (up to 5 years).
- Integrated **ClinicalBERT ICD code embeddings** with a **BEHRT sequence transformer** and age-aware positional encodings, achieving an **AUROC of 0.81** and outperforming **Mamba and LSTM baselines**.
- Evaluated **transformer attention weights** and **t-SNE/UMAP** projections to isolate early AMD precursor diagnoses for interpretable ophthalmology screening.

Zelta Automations - Algorithmic Trading

- Built and backtested **quantitative statistical arbitrage strategies for BTC and ETH** using technical trend filters and disciplined risk management at **Inter IIT Tech Meet 13.0**.
- Applied **Hawkes-process modeling for volatility estimation** and market regime analysis, designing a **multi-timeframe system** that aligned high-frequency market events with lower-frequency execution for **BTC/ETH pairs-trading**.
- Implemented **Kalman filtering to denoise price dynamics and smooth return series**, minimizing market noise while preserving signal conviction.

POSITIONS OF RESPONSIBILITY

Overall Head, Infero - *Competitive Programming and logical reasoning Club.*

- Spearheaded a 15+ member team to drive the Competitive Programming culture at IITH, organizing flagship events like **QuantFest** and **CodeFest**.
- Curated original problems and hosted **7 campus-wide contests** covering advanced DSA for placement and internship preparation.